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Codebook

Codebook 



Easy

General Skills

Beginner picoMini 2022

shell

Python

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Description

Run the Python script `code.py` in the same directory as `codebook.txt`.

- [Download code.py](#)
- [Download codebook.txt](#)

Hints 

1 2

On the webshell, use `ls` to see if both files are in the directory you are in

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Author: The Analyst: Hyposelenia

Challenge: Codebook

Category: General Skills

Date: 01/17/26



I. Objective

The objective of this challenge is to successfully run a provided Python script in the **same directory** as a required text file in order to retrieve the hidden flag.

II. Background

This challenge focuses on understanding how **files within the same directory interact**, particularly how a Python script accesses and reads data from a text file.

In Linux, programs rely on **file paths** to locate required resources. If a script and its dependent file are not placed in the same directory (or properly referenced), the program will fail to execute correctly. Rather than manually decrypting the contents, the challenge emphasizes correct file organization and execution.

The provided hint states that **reverse engineering the encryption function is unnecessary**, meaning the solution relies on proper execution rather than cryptographic analysis.

III. Tool Used

- **Oracle VirtualBox (Kali Linux)** – Used as the Linux environment
- **Linux Terminal** – Used to execute decoding commands
- **wget** – Used to download the Python script from the challenge server

IV. Methodology

1. Downloaded the provided .txt file from the challenge and placed it inside a dedicated directory used for CTF-related files.



2. Launched **Kali Linux** using Oracle VirtualBox.
3. Since I placed the txt file in that directory, I right click that directory.
4. Opened the **Linux terminal** from there.
5. Used the **wget** command to download the Python script directly into the same directory as the text file.

```
kali@kali: /media/sf_Downloaded_Cybersecfiles
$ wget https://artifacts.picoctf.net/c/1/code.py
2026-01-16 23:48:04 -- https://artifacts.picoctf.net/c/1/code.py
Resolving artifacts.picoctf.net (artifacts.picoctf.net)... 18.172.21.12, 18.172.21.43, 18.172.21.90, ...
Connecting to artifacts.picoctf.net (artifacts.picoctf.net)|18.172.21.12|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1278 (1.2K) [application/octet-stream]
Saving to: 'code.py'
code.py 100% [1278/1278] 1.25K --KB/s in 0.001s
2026-01-16 23:48:08 (951 KB/s) - 'code.py' saved [1278/1278]
```

6. Verified that both the Python script and the text file were in the same directory using the **ls** command.

```
2026-01-16 23:48:08 (951 KB/s) - 'code.py' saved [1278/1278]
Documents
(kali@kali)-[/media/sf_Downloaded_Cybersecfiles]
$ ls
challenge.zip codebook.txt code.py enc_flag
```

7. Executed the Python script using **python3 <python script name>**.

```
(kali@kali)-[/media/sf_Downloaded_Cybersecfiles]
$ python3 code.py
picoCTF{c0d3b00k_455157_d9aa2df2}
```

8. Observed the output produced by the script, which revealed the flag.
9. Retrieved the flag.

V. Result

picoCTF{c0d3b00k_455157_d9aa2df2}



```
picoCTF{c0d3b00k_455157_d9aa2df2}
```

VI. Explanation

The **Python script** requires access to the accompanying text file in order to function correctly. When both files are placed in the **same directory**, the script can locate and read the text file without needing an absolute or relative path.

If the files were placed in different directories, the script would fail with a file-not-found error because it would not be able to locate the required resource.

The `wget` command was used to download the Python script directly into the working directory, ensuring both files were properly connected. The `ls` command was used to confirm their presence before execution.

Additionally, the challenge hint explicitly states that the `str_xor` function does **not** need to be reverse engineered. This confirms that the intended solution is to correctly run the program rather than manually decrypting or modifying the code.

VII. Conclusion

This challenge demonstrates the importance of understanding **directory structure, file dependencies, and basic Linux commands** in CTF competitions. Rather than focusing on complex decoding techniques, the solution relied on correct file placement and execution. Mastery of these fundamentals is essential for solving more advanced challenges in future CTFs.

— The Analyst: Hyposelenia